SIMON JUPP

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EDUCATION

Yale University **New Haven, CT**
B.S. in Computer Science Expected May 2025
GPA 3.8/4.0
Coursework: Data Structures • Algorithms • Systems Programming & Computer Organization • Object Oriented Programming • Full Stack Web Development • Computational Intelligence for Games • Machine Learning • Building Game Engines

SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, SQL, R, Assembly, D
- **Frameworks/Tools:** TensorFlow, Keras, PyTorch, Django, React, NumPy, Git, CUDA, Matplotlib
- **Web Development:** HTML, CSS, Django, React
- **Data Science & ML:** Machine Learning, Computer Vision, NLP

WORK EXPERIENCE

Enrich Biosystems Inc. **Branford, CT**
Software Engineer Intern June 2024 - Aug. 2024
- Developed a machine learning pipeline to streamline the integration of new training data, incorporating a custom-built GUI to enhance usability and support scalable workflows in a dynamic startup environment
- Created and incorporated computer vision similarity algorithms to highlight specialized cancer and T-cell interactions.

Enrich Biosystems Inc. **Branford, CT**
Software Engineer Intern May 2023 - Dec 2023
- Developed and implemented deep learning image segmentation (ResNet, UNet) models using PyTorch and TensorFlow to enhance the accuracy of cell identification and classification algorithms.
- Conducted data preprocessing and cleaning to prepare large datasets for model training, including data augmentation and normalization techniques.
- Implemented an automated version control pipeline for Sphinx documentation website deployment upon new code releases.

Danbury Grassroots Academy **Danbury, CT**
Intern Jan. 2021 - July 2021
- Co-created a program that paired first-generation and low-income students with their own tutor during COVID times.
- Recruited and managed 26 math tutors.
- Tutored 20 elementary, middle school, and high school students in math.

PROJECTS

Yale Bulldog Buddies
- Created a full-stack app allowing students to find study partners based on a variety of user preferences.
- Developed the backend using **Django** and designed models for profile information.
- Created a **REST API** to allow calls from the frontend to fetch and put data based on user input.

Rhythm Game Engine
- Developed a custom game engine for a rhythm-based game using **D** and **SDL2**, implementing core game design patterns (e.g., observer, state, component) to manage game flow, input handling, and scene transitions.
- Engine includes real-time hit detection and scoring mechanics, allowing users to create, test, and refine custom levels using a modular, user-friendly level editor.

Task Manager
- Built a full-stack application to allow users to plan events and practice time management.
- Used **JSON databases** to simulate a backend and maintain the user's tasks upon restarting the application.
- Used **Javascript, React, CSS** on the frontend for a dynamic representation of their tasks.

LEADERSHIP & ACTIVITIES

Yale Varsity Track & Field **New Haven, CT**
Student Athlete Fall 2020 - Present
- Train 20 hours per week throughout the academic year while achieving top results in the Ivy League Conference.

Yale Student Athlete Advisory Committee **New Haven, CT**
Member Fall 2022 – Spring 2023
- Collaborated with university administrators to provide actionable insights on the student-athlete experience, influencing policy changes affecting athletic programs.
- Contributed input on new regulations to improve student-athlete welfare, ensuring alignment with NCAA and Ivy League guidelines.